The Fourier Transform

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Key Concepts

1) The Fourier transform is used to analyze signals and systems.

2) The Fourier transform applies to continuous-time non periodic signals, but can be extended to discrete-time and/or periodic signals using impulses.

3) Fourier transform

$$X(\Omega) = \int_{-\infty}^{\infty} x(t) e^{-j\Omega t} dt$$

4) Inverse Fourier transform

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\Omega) e^{j\Omega t} d\Omega$$

5) The inverse Fourier transform expresses $x(t)$ as a weighted "sum" of sinusoids of different frequencies.

6) The integrals in the Fourier transform do not converge for all signals.

7) Fourier transform pairs

a)

$$x(t) = e^{-at}u(t) \overset{FT}{\longleftrightarrow} X(\Omega) = \frac{1}{a + j\Omega}$$

b)

$$x(t) = \begin{cases} 1, & |t| < T_o \\ 0, & |t| > T_o \end{cases} \overset{FT}{\longleftrightarrow} X(\Omega) = \frac{2\sin(\Omega T_o)}{\Omega}$$

c)

$$x(t) = \frac{\sin(Wt)}{\pi t} \overset{FT}{\longleftrightarrow} X(\Omega) = \begin{cases} 1, & |\Omega| < W \\ 0, & |\Omega| > W \end{cases}$$

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d)

$$x(t) = \delta(t) \overset{FT}{\longleftrightarrow} X(\Omega) = 1$$

e)

$$x(t) = \frac{1}{2\pi} \overset{FT}{\longleftrightarrow} X(\Omega) = \delta(\Omega)$$

8) The width of a signal in time is inversely related to the width of its Fourier transform.

Video Screenshots

- Analysis of signals and systems in continuous and discrete time
- Continuous-time, nonperiodic signals. Extend to discrete-time and periodic

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\Omega) e^{j\Omega t} d\Omega \qquad X(\Omega) = \int_{-\infty}^{\infty} x(t) e^{-j\Omega t} dt$$

$$x(t) \overset{FT}{\longleftrightarrow} X(\Omega)$$

- Express $x(t)$ as weighted sum of sinusoids

$$x(t) \approx \sum_{i=-\infty}^{\infty} X(\Omega_i) \frac{\Delta\Omega}{2\pi} e^{j\Omega_i t}$$

- Convergence

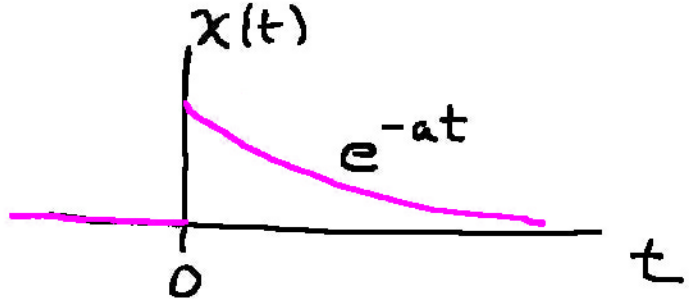
Example: Decaying Exponential

$x(t) = e^{-at}u(t)$

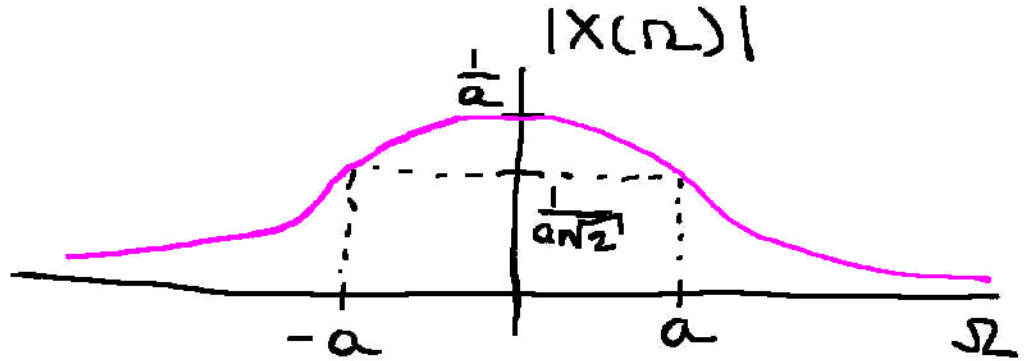
$$X(\Omega) = \int_{-\infty}^{\infty} e^{-at}u(t)\,e^{-j\Omega t}\,dt$$

$$= \int_{0}^{\infty} e^{-(a+j\Omega)t}\,dt \qquad (a>0)$$

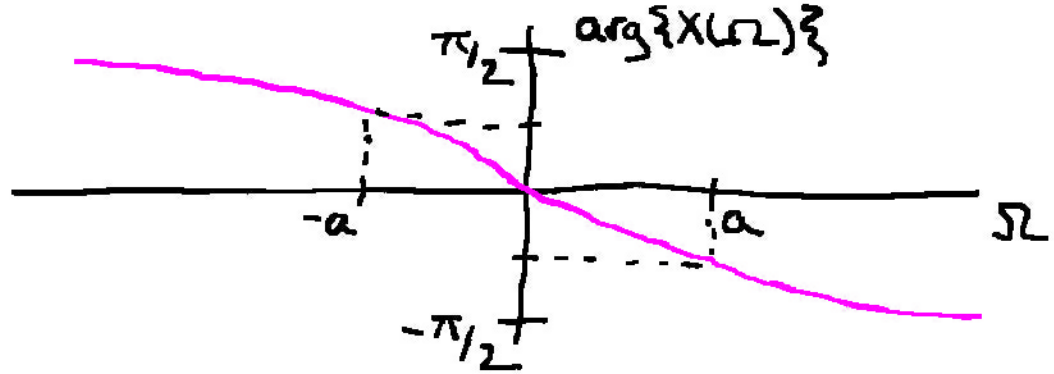
$$= \frac{-1}{a+j\Omega}\,e^{-(a+j\Omega)t}\Big|_{0}^{\infty}$$



$$X(\Omega) = \frac{1}{a+j\Omega}$$



$$|X(\Omega)| = \frac{1}{(a^2+\Omega^2)^{1/2}}$$



$$\arg\{X(\Omega)\} = -\arctan\left(\frac{\Omega}{a}\right)$$

Example: Rectangular Pulse

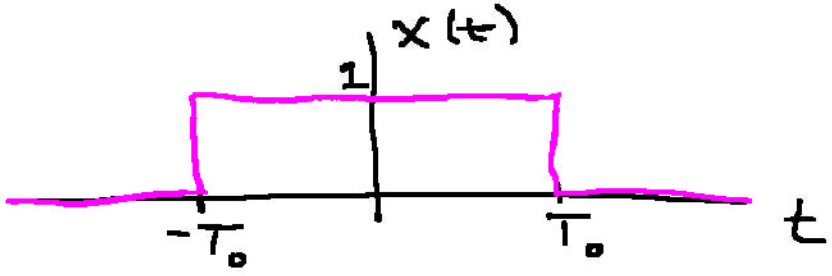
$x(t) = \begin{cases} 1 & |t| < T_0 \\ 0 & |t| > T_0 \end{cases}$

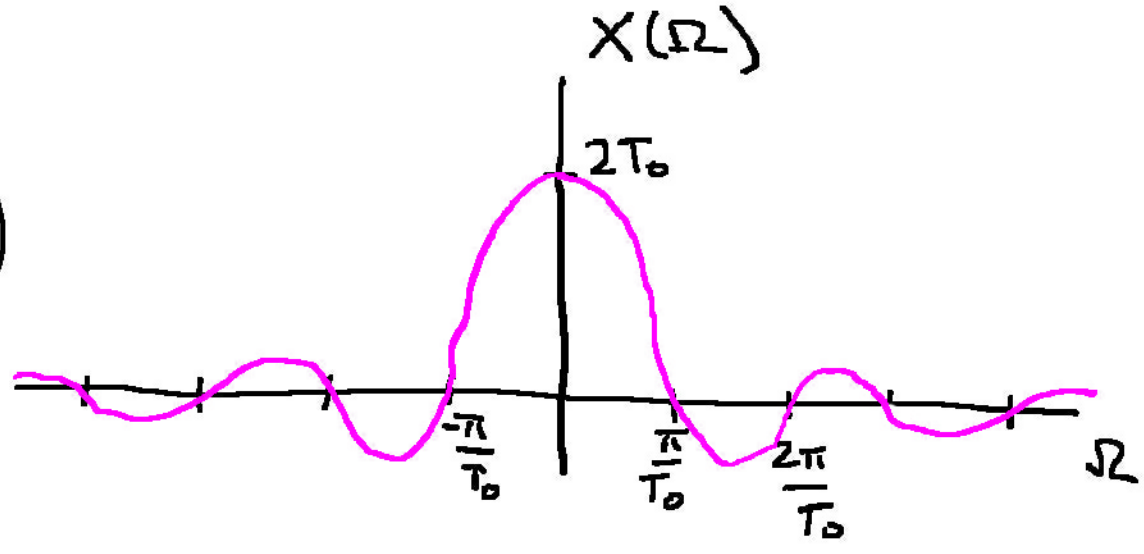
$$X(\Omega) = \int_{-T_0}^{T_0} 1 \cdot e^{-j\Omega t}\, dt$$

$$= \begin{cases} 2T_0 & \Omega = 0 \\ \underbrace{\frac{1}{j\Omega} e^{-j\Omega t}\Big|_{-T_0}^{T_0}}_{\frac{1}{j\Omega}\left(e^{-j\Omega T_0} - e^{j\Omega T_0}\right)} & \Omega \neq 0 \end{cases}$$

$$X(\Omega) = \frac{2 \sin \Omega T_0}{\Omega}$$

Note: $\lim_{\Omega \to 0} \frac{2 \sin \Omega T_0}{\Omega} = 2T_0$





"Width" in t and Ω is inversely related

Example: Rectangular Spectrum

$$X(\Omega) = \begin{cases} 1, & |\Omega| < W \\ 0, & |\Omega| > W \end{cases}$$

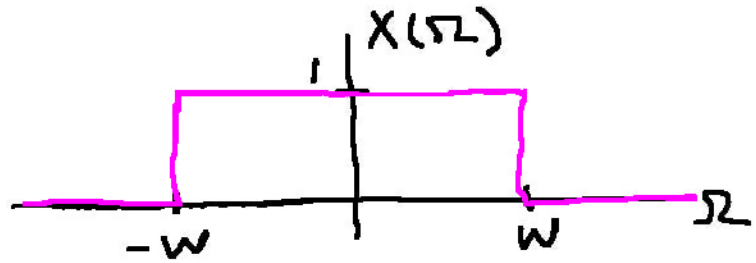
$$x(t) = \frac{1}{2\pi} \int_{-W}^{W} 1 \, e^{j\Omega t} \, d\Omega$$

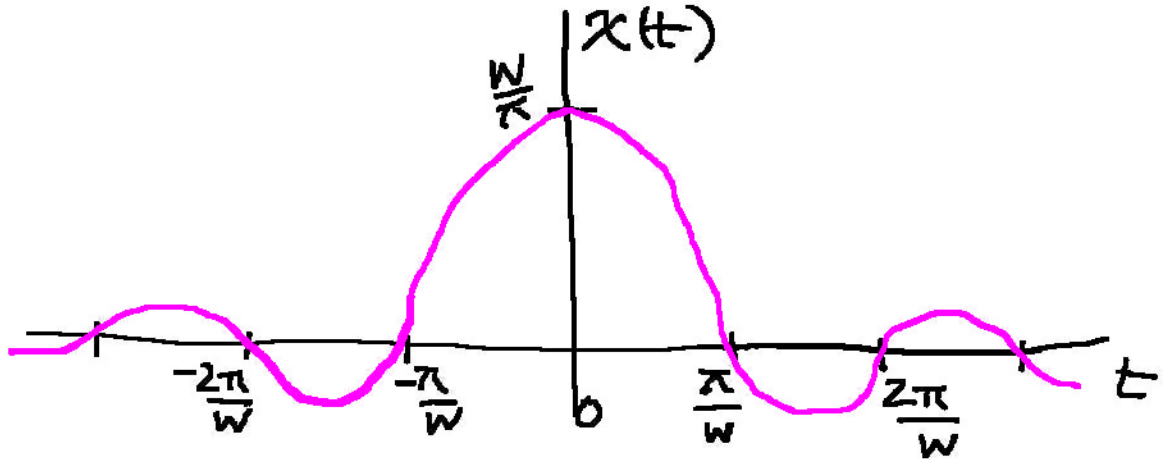
$$= \frac{1}{2\pi j t} \, e^{j\Omega t} \Big|_{-W}^{W} \quad (t \neq 0)$$

$$= \begin{cases} \dfrac{1}{\pi t} \sin(Wt) & t \neq 0 \\ W/\pi & t = 0 \end{cases}$$

$$x(t) = \frac{\sin(Wt)}{\pi t}$$

Note: $\lim_{t \to 0} \dfrac{\sin(Wt)}{\pi t} = \dfrac{W}{\pi}$



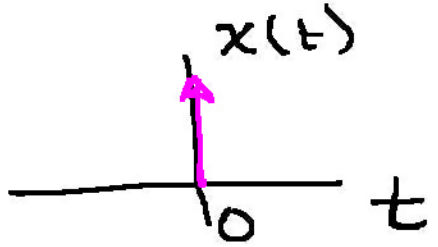


"Width" in t and Ω inversely related

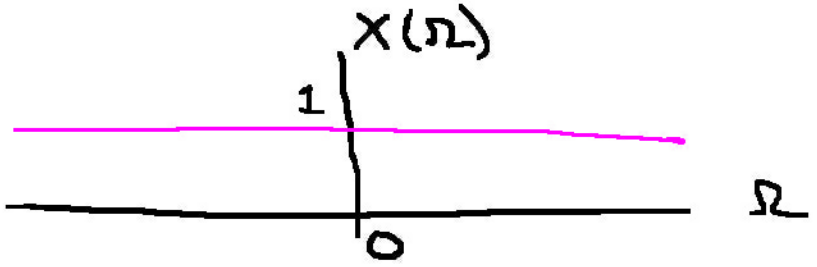
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Example: FT of Impulse

$x(t) = \delta(t)$



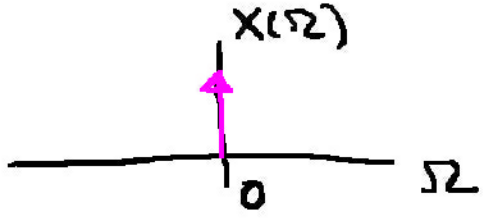
$$X(\Omega) = \int_{-\infty}^{\infty} \delta(t)\, e^{-j\Omega t}\, dt$$



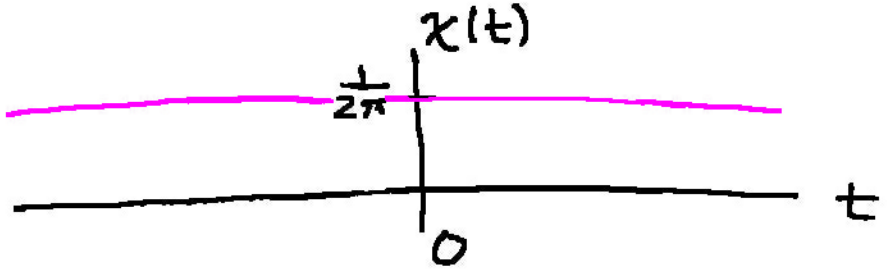
$$X(\Omega) = 1$$

Example: Inverse FT of Impulse

$X(\Omega) = \delta(\Omega)$



$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \delta(\Omega)\, e^{j\Omega t}\, d\Omega$$



$$x(t) = \frac{1}{2\pi}$$

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